

# 50 .NET Interview Questions & Answers

Prepared for LinkedIn lead generation and interview preparation.

## **1. What is .NET?**

.NET is Microsoft's development platform for building web, desktop, mobile, cloud, and IoT applications.

## **2. What is CLR?**

Common Language Runtime manages memory, exceptions, security, and execution of .NET applications.

## **3. What is CTS?**

Common Type System defines how data types are declared and used across .NET languages.

## **4. What is CLS?**

Common Language Specification provides language interoperability rules.

## **5. Difference between .NET Framework and .NET?**

.NET Framework is Windows-only; modern .NET is cross-platform.

## **6. What is managed code?**

Code executed under CLR supervision.

## **7. What is garbage collection?**

Automatic memory management that reclaims unused objects.

## **8. What is an assembly?**

A compiled .dll or .exe containing IL code and metadata.

## **9. What is IL?**

Intermediate Language generated before JIT compilation.

## **10. What is JIT?**

Just-In-Time compiler converts IL into machine code at runtime.

## **11. Value type vs Reference type?**

Value types store data directly; reference types store references.

## **12. What is boxing?**

Converting a value type to object/reference type.

## **13. What is unboxing?**

Converting boxed object back to value type.

## **14. What is the difference between string and StringBuilder?**

String is immutable; StringBuilder is mutable.

**15. What is dependency injection?**

Technique for supplying dependencies from external sources.

**16. What is ASP.NET Core?**

Cross-platform framework for building modern web applications and APIs.

**17. What is middleware?**

Components in the request processing pipeline.

**18. What is Kestrel?**

Cross-platform web server used by ASP.NET Core.

**19. What is appsettings.json?**

Configuration file for application settings.

**20. What is IConfiguration?**

Interface used to access configuration values.

**21. What is Entity Framework Core?**

ORM for interacting with databases using .NET objects.

**22. Code First vs Database First?**

Code First starts from classes; Database First starts from an existing DB.

**23. What is DbContext?**

Primary EF Core class for database operations.

**24. What is migration?**

Mechanism to evolve database schema over time.

**25. What is LINQ?**

Language Integrated Query for querying collections and data sources.

**26. IEnumerable vs IQueryable?**

IEnumerable executes in memory; IQueryable executes at data source level.

**27. First vsFirstOrDefault?**

First throws if none found; FirstOrDefault returns default value.

**28. What is async/await?**

Pattern for asynchronous programming.

**29. Task vs Thread?**

Task is a higher-level abstraction over threads.

### **30. What is REST API?**

Architectural style using HTTP methods for resources.

### **31. GET vs POST?**

GET retrieves data; POST creates resources.

### **32. PUT vs PATCH?**

PUT replaces resource; PATCH partially updates it.

### **33. What is JWT?**

JSON Web Token used for authentication and authorization.

### **34. Authentication vs Authorization?**

Authentication verifies identity; authorization controls access.

### **35. What is SignalR?**

Framework for real-time communication.

### **36. What is CORS?**

Cross-Origin Resource Sharing controls browser access across domains.

### **37. What is a delegate?**

Type-safe method reference.

### **38. What is an event?**

Delegate-based notification mechanism.

### **39. Abstract class vs Interface?**

Abstract class can contain implementation; interface defines contracts.

### **40. Method overloading?**

Same method name with different parameters.

### **41. Method overriding?**

Redefining inherited virtual method behavior.

### **42. What is encapsulation?**

Hiding internal implementation details.

### **43. What is inheritance?**

Acquiring properties and behavior from another class.

### **44. What is polymorphism?**

Ability to use one interface for different implementations.

### **45. What is SOLID?**

Five principles for maintainable object-oriented design.

***46. What is Singleton pattern?***

Ensures only one instance exists.

***47. Transient vs Scoped vs Singleton?***

DI lifetimes controlling object creation.

***48. What is Azure Functions?***

Serverless compute service for event-driven workloads.

***49. What is Blazor?***

Framework for building interactive web UIs using C#.

***50. What is Cosmos DB?***

Globally distributed NoSQL database service from Azure.